Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A18I · Group N

Report date 31 Oct 2026

Group N · No progression recorded Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**53**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**CM0 (I+)**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

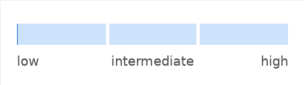
Sample: **TCGA-BH-A18I-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.83** · MSIsensor score: **0.04** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

0th

Low risk tier · fused neuro-symbolic model



Probability of progression

5% by 36 mo

12 mo 1% · 24 mo 3% · 60 mo 11%

Model-estimated progression time

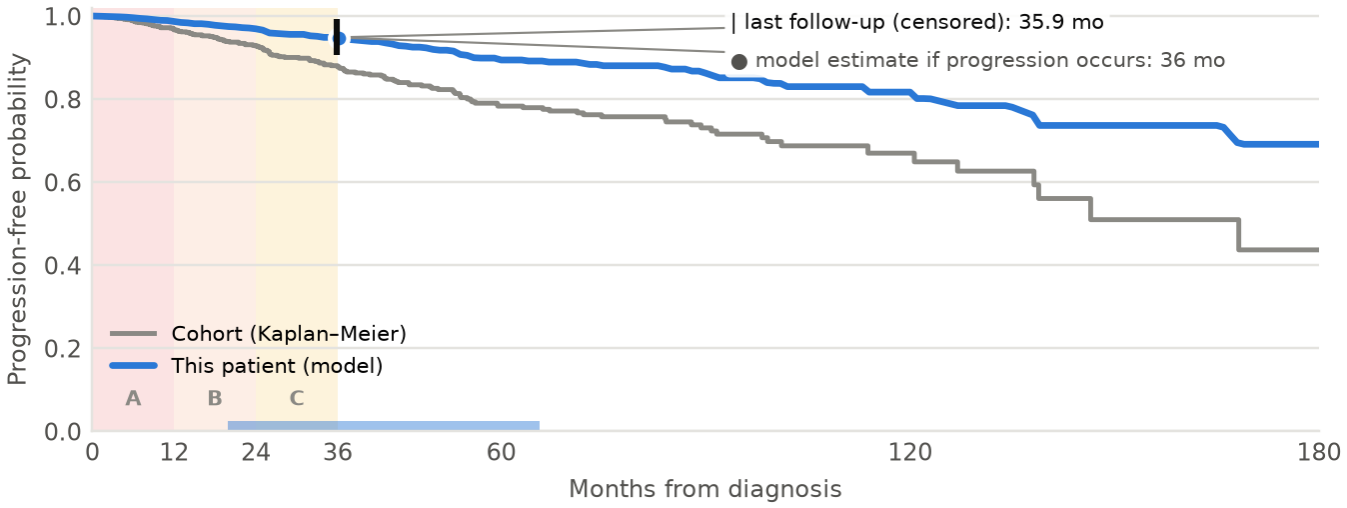
≈ 36 months

Window > 36 months · IQR 20–66 mo if progression occurs

Recorded progression

None recorded

Followed 35.9 mo · **Group N** · No progression recorded
Censored at last follow-up; not proof of lifelong remission.



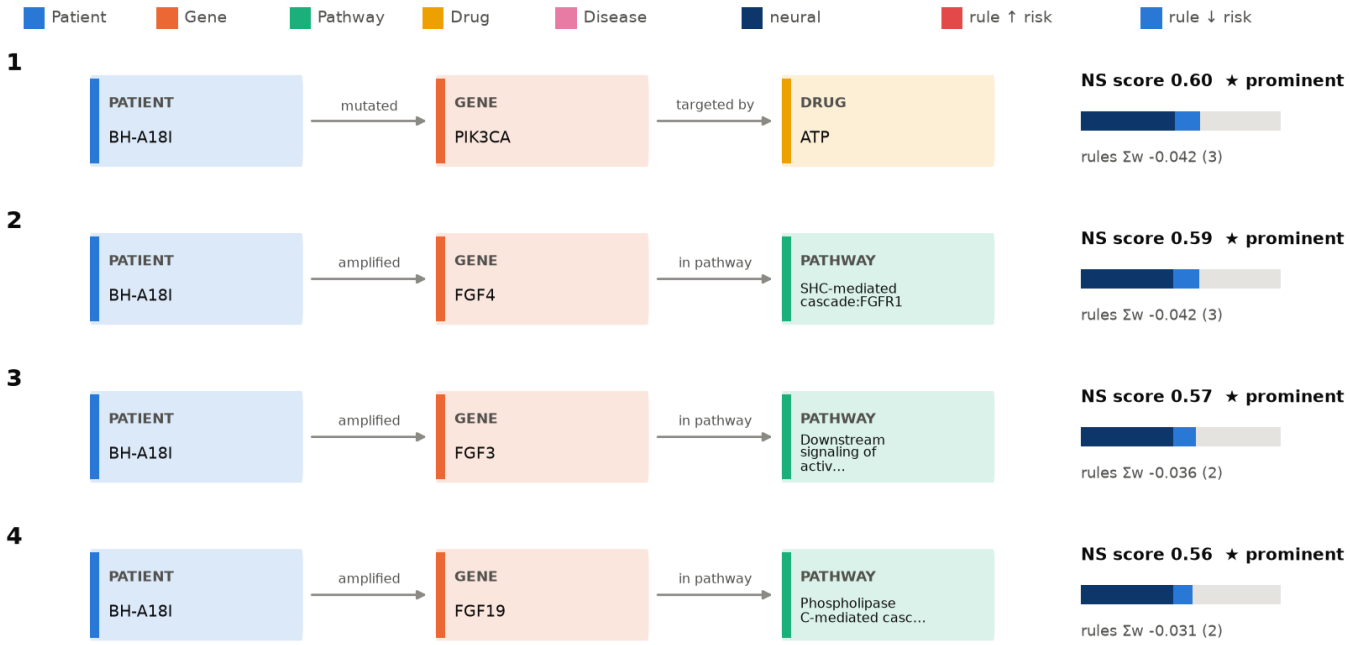
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 53-year-old patient has a LumA tumour, AJCC stage II (T1, N1, M0). The fused neuro-symbolic model places them in the **0th risk percentile** (low tier). The strongest learned formula that holds for them (*TStage(p, T1)* ∧ *Subtype(p, LumA)* ⇒ *Progress(p)*) lowers the progression hazard (weight -0.036 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **ATP**; the learned formulas it instantiates point to slower progression (Σw -0.042). No progression was recorded during 35.9 mo of follow-up.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

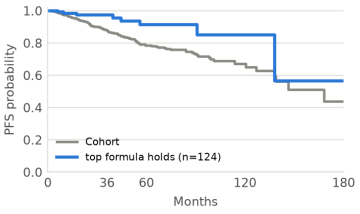


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A18I** mutated → **PIK3CA** targeted by → **ATP**	78	↓ **slower progression** · Σw -0.042 over 3 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.70, n = 339 carriers	**0.60**
2 ★	**TCGA-BH-A18I** amplified → **FGF4** in pathway → **SHC-mediated cascade:FGFR1**	77	↓ **slower progression** · Σw -0.042 over 3 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.91, n = 207 carriers	**0.59**
3 ★	**TCGA-BH-A18I** amplified → **FGF3** in pathway → **Downstream signaling of activated FGFR1**	77	↓ **slower progression** · Σw -0.036 over 2 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.91, n = 207 carriers	**0.57**
4 ★	**TCGA-BH-A18I** amplified → **FGF19** in pathway → **Phospholipase C-mediated cascade; FGFR4**	77	↓ **slower progression** · Σw -0.031 over 2 formula(s) Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) cohort: HR 0.73, n = 127 carriers	**0.56**

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

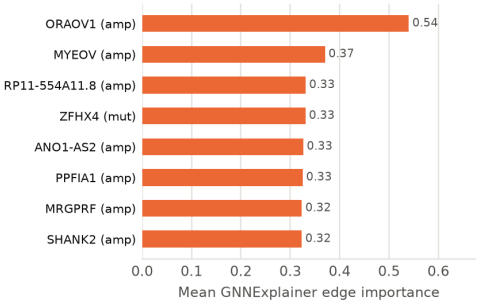
Weight	Formula (first-order, grounded on this patient)	Source	Support
-0.036 ↓ risk	TStage(p, T1) ∧ Subtype(p, LumA) ⇒ Progress(p)	mined pair	124 pts
-0.025 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling') ⇒ Progress(p) [KG]	mined pair	320 pts
-0.011 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR2') ⇒ Progress(p) [KG]	mined pair	233 pts
-0.006 ↓ risk	Stage(p, II) ∧ Altered(p, g) ∧ InPathway(g, 'PI-3K cascade:FGFR4') ⇒ Progress(p) [KG]	mined pair	233 pts



Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.E542K	70	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
ORAOV1	High-level amplification	GISTIC ±2	0	0.54
MYEOV	High-level amplification	GISTIC ±2	0	0.37
RP11-554A11.8	High-level amplification	GISTIC ±2	0	0.33
ZFHX4	Mutation	p.I3222Nfs*14	0	0.33
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.33
PPFIA1	High-level amplification	GISTIC ±2	6	0.33
MRGPRF	High-level amplification	GISTIC ±2	0	0.32
SHANK2	High-level amplification	GISTIC ±2	1	0.32
SHANK2-AS1	High-level amplification	GISTIC ±2	0	0.32
ANO1	High-level amplification	GISTIC ±2	2	0.32



20 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718) … and 6 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0BD	47	II	LumB	0.90	no progression, followed 18.2 mo
TCGA-BH-A0W3	58	II	LumB	0.90	no progression, followed 23.9 mo
TCGA-C8-A12Y	44	II	LumA	0.88	no progression, followed 48.5 mo
TCGA-E2-A15K	58	II	LumB	0.84	no progression, followed 9.0 mo
TCGA-AO-A126	39	II	LumA	0.83	progressed at 8.8 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.